



BACKDROPS · SPRITES

Create the space environment and add a spaceship



STEPS TO FOLLOW

1. Open scratch.mit.edu and click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" → search "Stars" black starfield
4. Click "Choose a Sprite" → search "Rocketship" → add it
5. Set size to 60 → drag the ship to the bottom-centre

TIP

Set the rocket's direction to 0°

(pointing up) in Sprite Info so the
rocket faces upward.

DID YOU KNOW?

The Scratch stage is 480×360 pixels.

$x=0$, $y=0$ is the centre.



EVENTS · MOTION · CONTROL

Move the spaceship left and right with arrow keys



SCRATCH BLOCKS — ON THE ROCKETSHIP SPRITE

when  clicked

go to x: (0) y: (-150)

forever

if <key [left arrow] pressed?> then

change x by (-8)

if <key [right arrow] pressed?> then

change x by (8)

if on edge, bounce

TIP

Test now — the ship should glide left and right without flying off screen.

DID YOU KNOW?

"if on edge, bounce" saves you from writing edge-detection code yourself.



SPRITE TEMPLATES · HIDING

Add a Ball sprite and hide it on start — ready to become a laser clone



SCRATCH BLOCKS — ON THE BALL SPRITE

On *BALL*: when  clicked

hide

TIP

Any sprite that spawns clones must hide its original first — otherwise the "master" copy sits frozen in the middle of the stage.

DID YOU KNOW?

This template-and-clone pattern is identical to the "class and instance" concept in OOP. The Ball sprite is the class; each clone is an instance.



OBJECT INSTANTIATION · CLONE PATTERN

Press **SPACE** to spawn a Ball clone that appears at the ship's position



SCRATCH BLOCKS — ON BALL & ROCKETSHIP

On BALL: when I start as a clone

go to [Rocketship] ← snaps to ship first

show

On ROCKETSHIP: when [space] key pressed

create clone of [Ball]

TIP

"go to [Rocketship]" must be the FIRST block in the clone script — before show. Otherwise the clone appears at the wrong position!

DID YOU KNOW?

"create clone of [Ball]" is identical to writing "new Bullet()" in Java or Python. You are doing Object-Oriented Programming right now.



REPEAT-UNTIL · DESTRUCTION PATTERN

The clone shoots upward and deletes itself when it leaves the screen



SCRATCH BLOCKS — ON THE BALL SPRITE

On *BALL*: when I start as a clone

go to [Rocketship]

show

repeat until <(y position) > (175)>

change y by (15)

delete this clone

TIP

Increase "change y by (15)" to 20 or 25 for faster bullets. Faster bullets make the game easier!

DID YOU KNOW?

"delete this clone" frees memory — just like garbage collection in Python and Java. Without it, hundreds of off-screen bullets would pile up.



CLONES · TIMERS · PARALLEL EXECUTION

Aliens appear randomly at the top and descend toward the ship



SCRATCH BLOCKS — ON THE ALIEN SPRITE

ALIEN: when  clicked

hide

forever

wait (1.5) seconds

go to x: (pick random (-200) to (200)) y: (180)

create clone of [myself]

ALIEN: when I start as a clone

show

repeat until <(y position) < (-180)>

change y by (-3)

delete this clone

TIP

Lower the wait time (e.g. 0.8 secs) to spawn aliens faster. Higher "change y by" makes them fall quicker.

DID YOU KNOW?

Two "when  clicked" scripts on one sprite run at the same time. In real code, this is called concurrency or async programming.



SENSING · COLLISION DETECTION

Make an alien react when a laser touches it — play a sound as feedback



SCRATCH BLOCKS — ON THE ALIEN SPRITE

ALIEN: when I start as a clone

show

repeat until <(y position) < (-180)>

change y by (-3)

if <touching [Ball]?> then

play sound [pop] until done

delete this clone

TIP

Click the Sounds tab on the Alien sprite to add new sounds. "Zap", "Laser1", and "Pop" all feel great for hits.

DID YOU KNOW?

"touching [Ball]?" polls for overlap 30 times per second. Game engines like Unity call this an AABB check — behind every game's hit detection.



VARIABLES • REWARD STATE

Create a Score variable and add 10 points every time an alien is destroyed



SCRATCH BLOCKS — ON THE ALIEN SPRITE

ALIEN (inside repeat-until loop)

```
if <touching [Ball]?> then
```

```
  change [Score] by (10)
```

```
  delete this clone
```

TIP

Right-click the Score display on the stage and choose "Large readout" for a big, satisfying counter!

DID YOU KNOW?

Every leaderboard, achievement, and in-game economy is built on this pattern: event detected → variable updated → UI refreshed.



STATE MACHINE · PROGRAM TERMINATION

Lose a life when an alien escapes; the game ends when lives reach zero



SCRATCH BLOCKS — ON THE ALIEN SPRITE

(outside the repeat-until, after it ends)

change [Lives] by (-1)

if <(Lives) = (0)> then

say [Mission Failed! 💣] for (2) seconds

stop [all]

delete this clone

💡 TIP

Share your game by clicking "Share" at the top of Scratch — anyone with the link can play it!

🧠 DID YOU KNOW?

Traffic lights, login flows, vending machines, and game modes are all state machines. "PLAYING → GAME_OVER" is the simplest version.